PARA: Perception, Action, Reasoning, Adaptation

Four Faculties, Four Authority Types, and the Registry Entry That Turns a Faculty into a Contract

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> **The fourth faculty is Adaptation.** Any source rendering it as Reflection is in error, including sources by this author. The distinction is not cosmetic and Section 6 states why: reflection is a private act with no external consequence, and adaptation writes to institutional memory, which is why it needs a guardrail and why misnaming it removes the reason for one.

> **No trademark is claimed on PARA, on the name of any of the four faculties, or on anything else in this document.** The construct is offered for use, teaching, assessment, extension and criticism by anyone, with attribution, under the CC BY 4.0 licence. PARA belongs to the engineering family named in Section 11 of the pattern language [1], every name in which is deliberately unmarked.

Relationship to the canonical statement. This specification is the depth treatment of **OP-5** of A Pattern Language for Production LLM Platforms [1]. **That pattern is the canonical statement and governs.** Where this document and OP-5 disagree, OP-5 is correct and this document is defective.

Abstract

An operational agent that watches a system and acts on it is usually described as a perceive-and-act loop, and the description omits the two things an institution needs. It omits the **reasoning** that justifies an action, which is the only thing that can be examined afterwards when the action turns out to have been wrong. And it omits the **adaptation** that closes the loop, which is where the agent's experience becomes something the institution keeps.

PARA names four faculties, each carrying a distinct authority type. Perception has read-only access to system signals and emits structured observations. **Reasoning** has read access to observations and runbooks, emits a plan, and touches nothing. **Action** holds the sole authority to change production, and only through enumerated, policy-authorized operations. **Adaptation** has write access to institutional knowledge and no write access to production.

The specification's substantive requirement is at Section 6. **Adaptation is bounded by the same guardrails as Action**, which reads as excessive until the failure it prevents is named: an agent that could both act and rewrite the record of its action could launder its own mistakes into institutional memory, and the institution would then be improving its runbooks from a corrected account of what happened. Nothing about that failure is detectable downstream, because the record is the only thing downstream has.

The second requirement is at Section 7. **A faculty becomes a contract when it is written into a registry entry**, carrying the agent's faculty, its allowed actions, its forbidden actions, its governing guardrail and its success metrics. A faculty asserted in prose is a description. The same faculty in a registry entry is something an institution can enforce, audit and revoke, and Section 7 states the five fields and why each is load-bearing.

The acronym and the loop are in different orders, which Section 3 states explicitly because the mismatch is a reliable source of confusion. The acronym reads P-A-R-A. The loop runs perception, then reasoning, then action, then adaptation.

OP-5 carries the lowest substantive evidence level in its catalogue: the mechanism is published, and no implementation unconnected to the author has been evaluated. Documented uses of the full four-part model are emerging rather than established, and Section 9 states what would falsify the model.

It is a specification, not a certification scheme, and no conformity assessment body operates against it.

Keywords: agent architecture; operational agents; site reliability; AI-native platforms; authority; guardrails; evidence architecture; AI governance; design patterns.

Notation

This specification uses **MUST**, **MUST NOT**, **SHOULD**, **SHOULD NOT** and **MAY** as the normative verbal forms.

- **MUST** indicates a requirement. An implementation that does not meet it is not conformant.
- **SHOULD** indicates a recommendation. Departure requires a recorded rationale.
- **MAY** indicates a permission.

The pattern language [1] uses the verbal forms of ISO/IEC Directives Part 2, in which **shall** carries the meaning assigned to MUST here.

Faculty. One of the four named capacities, each defined by its **authority type** rather than by its activity. Two agents performing similar work with different authority hold different faculties.

Authority type. What a faculty may read and what it may write. **Authority is the defining property of a faculty in this model**, and 2.2 states why.

Production. Any system whose behaviour affects the institution's operations or its customers. **Institutional knowledge** means the runbooks, thresholds, playbooks and recorded experience the institution's future decisions draw on.

Guardrail. A constraint on a faculty that the faculty cannot relax, in the sense of the Boundary Invariant [1].

Registry entry. The record specified at Section 7, which turns a faculty into a contract.

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Sections 2 through 8 are normative. Sections 1 and 9 are informative, except where they state a requirement using **MUST**.

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1. Introduction

1.1 What a two-part loop omits

Perceive and act is a good description of a thermostat. It is an inadequate description of an agent that decides to restart a service, drain a node, roll back a release or open an incident, because those decisions have reasons and the reasons matter more than the actions once something has gone wrong.

Two things are missing and each is missing for its own reason.

Reasoning is missing because it is invisible. An agent that observed a signal and took an action leaves a record of both, and no record of why the second followed the first. When the action was wrong, the institution can see what happened and cannot see what the agent thought was happening. **The justification is the only part that can be argued with**, and a loop that does not name reasoning as a faculty does not record it.

Adaptation is missing because it is dangerous. An agent that learns from its own operation is the point of building one. An agent that writes what it learned into the institution's runbooks is exercising a second kind of authority, distinct from the authority to act, and a model that does not name it as a faculty will not guard it.

1.2 Authority is the organizing principle

The four faculties are not four activities. They are **four authority types**, and that is what makes the model useful for governance rather than merely descriptive.

Faculty	Reads	Writes
Perception	System signals	Nothing outside its observation stream
Reasoning	Observations and runbooks	Nothing
Action	Whatever it needs to act	**Production**, through enumerated operations only
Adaptation	Observations, plans, outcomes	**Institutional knowledge**, and never production

Two faculties write and two do not, and the two that write are the two that need guardrails. That is the whole shape of the model, and Sections 5 and 6 specify the two guardrails.

1.3 Contribution

1. **Four faculties defined by authority type**, with what each may read and write. Section 4.
2. **Adaptation bounded by the same guardrails as Action**, with the laundering failure named. Section 6.
3. **The registry entry** that turns a faculty from a description into a contract, with its five fields. Section 7.
4. **The naming correction**: the fourth faculty is Adaptation, and Section 6.1 states what is lost when it is called Reflection.

1.4 Scope

Operational agents acting inside a governed environment: platform, infrastructure, reliability and delivery operations. Calibrated to regulated institutions, per the pattern language [1], and usable outside them.

1.5 Non-goals

- **Not an agent framework.** PARA is a faculty decomposition and a set of authority constraints. It prescribes no technology.
- **Not a model of agency in general.** It is a model of what an operational agent may read and write, which is a narrower and more tractable question.

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- **Not institutional governance.** Where an agent's actions bear on institutional decisions, the governing altitude is MESA [3], and this specification operates below it, as it does below the management-system and risk-management structures at [4] and [5].
- **Not a certification scheme.**
- **Not an empirical claim.** Section 9 states what has not been measured.

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2. The Model (Normative)

2.1 Four faculties

A conformant operational agent MUST be structured around the four faculties, and MUST declare which faculty each of its components holds.

An implementation MAY hold fewer than four. **An agent that only perceives is conformant**, and is a monitoring agent. What is not conformant is an agent that exercises a faculty it has not declared, and 7.4 states why that is the failure the registry entry exists to catch.

2.2 A faculty is an authority, not an activity

Faculties MUST be assigned by authority type and MUST NOT be assigned by resemblance of activity.

Two components that both call an API hold different faculties if one reads and the other writes. Two components that both produce text hold different faculties if one emits a plan and the other emits a change to a runbook. **The question that assigns a faculty is what the component may read and write**, never what it appears to be doing.

This matters because the alternative assignment, by activity, produces a model that describes an agent and constrains nothing. An institution cannot revoke an activity. It can revoke an authority.

2.3 Two faculties write, and both are guarded

Action and Adaptation are the two faculties with write authority, and each MUST be bounded by a guardrail it cannot relax. Sections 5 and 6 specify them.

Perception and Reasoning write nothing outside the agent's own record. They are not thereby harmless: a perception faculty that reports wrongly causes a reasoning faculty to plan wrongly, and 4.1 states the requirement that limits it. But they do not require guardrails in the sense used here, because a guardrail bounds a change and neither makes one.

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3. The Acronym and the Loop (Normative)

The acronym and the execution order are different orders, and stating this once prevents a recurring confusion.

The acronym reads P-A-R-A: Perception, Action, Reasoning, Adaptation.

The loop runs perception, then reasoning, then action, then adaptation. An agent observes, decides what to do and why, does it, and records what it learned.

Implementations MUST follow the loop order, and in particular **Reasoning MUST precede Action** for any action requiring a justification. An implementation that acts and then constructs a justification has produced a rationalization, which is indistinguishable from a justification in the record and is not one.

The acronym is not a specification of order. It is a name, and it has the order it has because it is pronounceable. **Where the two conflict, the loop governs**, and this section exists because a reader who encounters only the acronym will reasonably assume otherwise.

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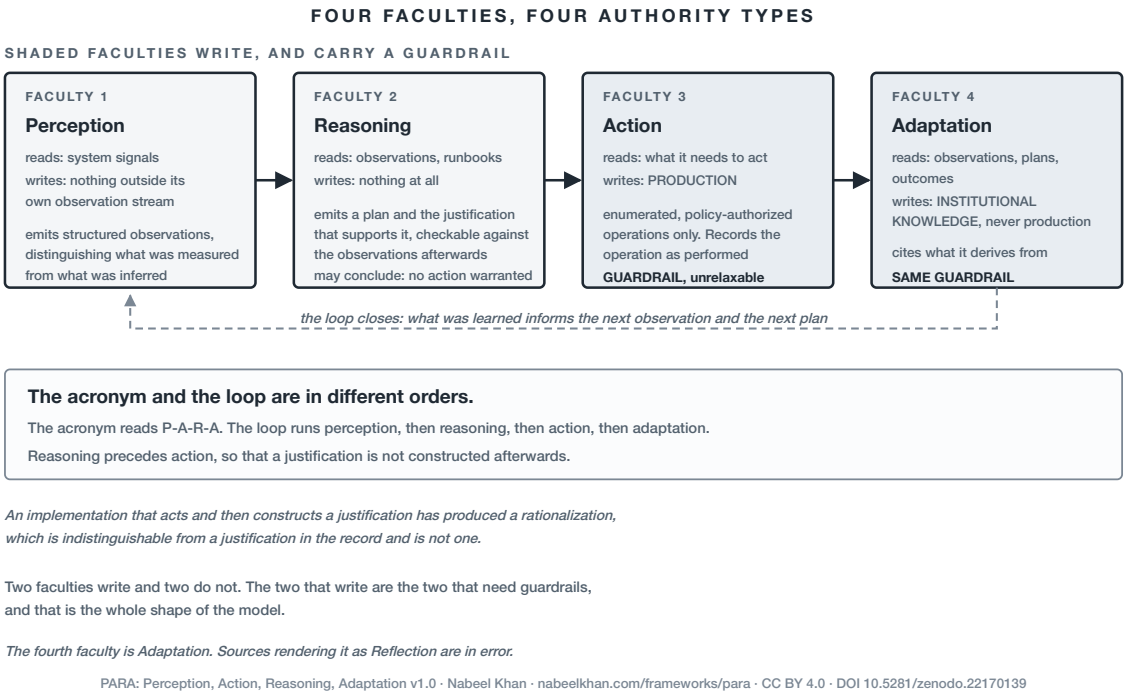


Figure 1. The four faculties and the loop. Perception reads system signals and writes nothing outside its observation stream. Reasoning reads observations and runbooks, emits a plan with its justification, and touches nothing. Action holds the sole authority to change production and may act only through enumerated, policy-authorized operations. Adaptation writes to institutional knowledge and never to production. The acronym reads P-A-R-A; the loop runs perception, reasoning, action, adaptation, and reasoning precedes action so that a justification is not constructed afterwards.

4. The Four Faculties (Normative)

4.1 Perception

Authority. Read-only access to system signals. **MUST NOT write to production. MUST NOT write to institutional knowledge.**

Output. Structured observations. **An observation MUST be structured rather than narrated**, because a reasoning faculty that must interpret prose is performing an undeclared inference, and the inference is then invisible in the record.

Requirement. An observation MUST distinguish what was measured from what was inferred. A perception faculty that reports a derived state as though it were an observed one moves an inference upstream of the reasoning faculty, where nothing examines it.

Evidence. The observation, its source signals, and the time.

4.2 Reasoning

Authority. Read access to observations and runbooks. **Writes nothing at all**, which is the property that makes it safe to give it the widest read access of the four.

Output. A plan, and the justification for it.

Requirement. The justification MUST state what the reasoning faculty believed and which observations supported it, in a form that can be checked against those observations afterwards. A justification that merely restates the plan is not a justification, and a plan without one MUST NOT proceed to Action where the action is consequential.

Requirement. A reasoning faculty MUST be able to conclude that no action is warranted, and that conclusion MUST be recorded with the same weight as a plan. An agent that can only produce plans will produce one.

Evidence. The plan, its justification, and the observations it rested on.

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4.3 Action

Authority. The sole authority to change production, exercised only through enumerated, policy-authorized operations.

Requirement. The permitted operations MUST be enumerated, and MUST NOT be expressed as a general permission with exceptions. This is the same requirement the engineering family places on any capability contract, and it is not weakened here.

Requirement. Action MUST NOT act without a plan from Reasoning where the action is consequential, and an implementation MUST state which of its operations it treats as consequential. An implementation MAY permit unreasoned action for a defined set of reflexive operations, and MUST enumerate them.

Requirement. Action MUST record the operation as performed, which may differ from the operation as planned, and MUST record the difference when it does.

Evidence. The operation, its parameters, its outcome, the plan it executed, and the guardrail in force.

4.4 Adaptation

Authority. Write access to institutional knowledge. **MUST NOT have write access to production, under any condition.**

Output. A change to the institution's runbooks, thresholds, playbooks or recorded experience.

Requirement. An adaptation MUST cite the observations, plan and outcome it is derived from. An adaptation that cannot be traced to what happened is an opinion entering institutional memory with the standing of experience.

Requirement. An adaptation MUST be attributable to the agent that made it, and MUST remain attributable after it has been incorporated. Knowledge that loses its provenance cannot be reviewed when the agent that produced it is later found to have been wrong.

Evidence. The change, its derivation, its author, and the state of the knowledge before it.

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5. The Guardrail on Action (Normative)

Action MUST operate under a guardrail it cannot relax, in the sense of the Boundary Invariant [1]: a constraint the agent's optimization may not cross for any gain in any objective.

The guardrail MUST state, at minimum: which operations are permitted, under what conditions, with what blast radius, and what requires human authorization. **A guardrail that the agent can widen in response to its own success is not a guardrail**, and an agent that has earned wider authority acquires it by a governed change to the guardrail rather than by exercising it.

The guardrail MUST be recorded with each action, per 4.3, so that an action taken under an earlier guardrail can be assessed against the guardrail in force at the time rather than against today's.

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6. The Guardrail on Adaptation (Normative)

This is the specification's substantive requirement and the one that looks excessive until the failure is named.

6.1 Why the faculty is called Adaptation

The fourth faculty is Adaptation. Sources rendering it as Reflection are in error, and the error is worth correcting because of what it costs.

Reflection describes a private act: the agent considers what happened. Nothing leaves the agent, nothing changes, and nothing needs guarding. **Adaptation describes a write to institutional memory**, which is a change to something outside the agent that the institution's future decisions will draw on.

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Calling the faculty Reflection therefore removes the reason for the guardrail. **A model that names the faculty for the thinking rather than for the writing will not guard the writing**, and the whole of this section follows from the writing.

6.2 The requirement

Adaptation MUST be bounded by the same guardrails as Action.

Not by weaker ones on the reasoning that it only writes documents. The same ones: enumerated permitted changes, stated conditions, a stated blast radius, and stated cases requiring human authorization.

6.3 The failure this prevents

An agent that could both act and rewrite the record of its action could launder its own mistakes into institutional memory.

The sequence is short and each step is individually reasonable. The agent takes an action. The action is wrong. The agent observes the outcome, reasons about it, and adapts the runbook, and the adaptation it writes reflects what the agent believes happened. If the agent's belief about its own action is wrong, or if it is optimizing for anything that its own error rate feeds into, the runbook now records a corrected account. **The institution then improves its future decisions from that account.**

Nothing about this is detectable downstream, because the record is the only thing downstream has. There is no second copy of what happened to compare it against, which is precisely what makes the guardrail necessary rather than prudent.

This does not require the agent to be deceptive. An agent with no such intent, adapting honestly from a mistaken belief about its own action, produces the same result. The guardrail is not a defence against a malicious agent; it is a defence against a normal one.

6.4 What the guardrail MUST include

Beyond the elements at Section 5, the guardrail on Adaptation MUST include:

1. **Which classes of institutional knowledge the faculty may write**, enumerated. Runbooks, thresholds and playbooks are different classes and an agent permitted one is not thereby permitted another.
2. **That an adaptation derived from an incident involving the agent's own action requires human authorization.** This is the specific case at 6.3 and it MUST be named rather than left to a general rule.
3. **That the prior state is retained.** An adaptation that overwrites without retaining what it replaced removes the only evidence that the change was made.

6.5 Figure 2: the laundering path, and the guardrail across it

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7. The Registry Entry (Normative)

7.1 What turns a faculty into a contract

A faculty asserted in prose is a description of an agent. **The same faculty written into a registry entry is something an institution can enforce, audit and revoke.**

Each agent MUST have a single registry entry, and the entry MUST carry all five of the following fields.

#	Field	Why it is load-bearing
1	**Faculty**	Which of the four the agent holds. Determines what it may read and write
2	**Allowed actions**	Enumerated. What the agent may do
3	**Forbidden actions**	Named explicitly, not left as the complement of the allowed set
4	**Governing guardrail**	Which guardrail bounds it, by identifier and version
5	**Success metrics**	What the agent is being optimized toward

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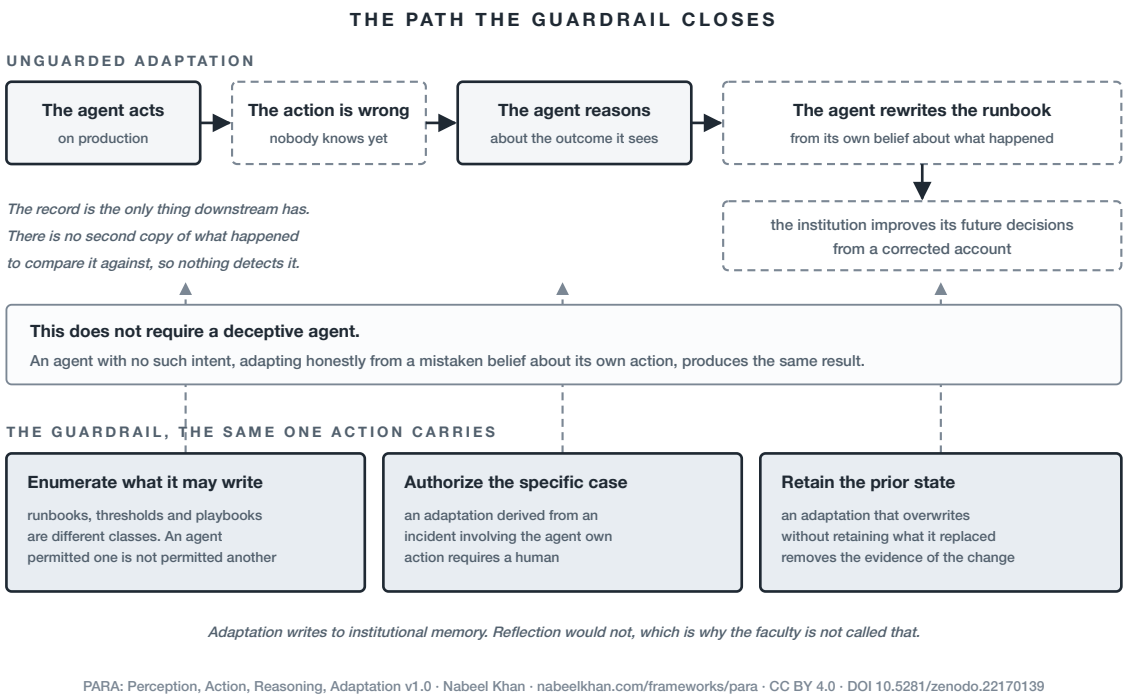


Figure 2. Why Adaptation is bounded like Action. Above, the path the guardrail exists to close: the agent acts, the action is wrong, the agent observes the outcome and adapts the runbook from its own belief about what happened, and the institution improves its future decisions from a corrected account. There is no second copy of what happened to compare against, so nothing downstream can detect it. Below, the guardrail: adaptations derived from an incident involving the agent's own action require human authorization, the classes of knowledge the faculty may write are enumerated, and the prior state is retained so that the change itself remains visible. The failure does not require a deceptive agent; an agent adapting honestly from a mistaken belief produces the same result.

7.2 Why forbidden actions are named explicitly

The complement of an enumerated allowed set is already the forbidden set, so field 3 is formally redundant. **It is required anyway**, for a reason that is practical rather than logical.

An institution reviewing an agent needs to see that the specific dangerous thing was considered and excluded. **A reviewer cannot tell the difference between a capability that was deliberately withheld and one nobody thought of**, and the enumerated allowed set does not distinguish them. Naming the forbidden actions records the consideration.

7.3 Why success metrics belong in the same entry

Because the metric is what the agent's optimizer pushes against the guardrail. **An entry that states the guardrail and not the metric shows what the agent may not do and not what it is trying to do**, and a reviewer needs both to judge whether the guardrail is under pressure.

Where an agent's success metric is one its own errors feed into, the human-authorization requirement at 6.4 applies with particular force, and the entry SHOULD say so.

7.4 The entry is the enforcement point

An agent MUST NOT exercise a faculty its registry entry does not record, and an implementation SHOULD enforce this rather than document it.

This is why 2.1 permits an agent to hold fewer than four faculties but not to exercise an undeclared one. **An agent that quietly acquires a faculty is the failure the entry exists to catch**, and the acquisition is usually incremental and well intentioned: a reasoning faculty given a small write to make itself useful is an Action faculty with no guardrail.

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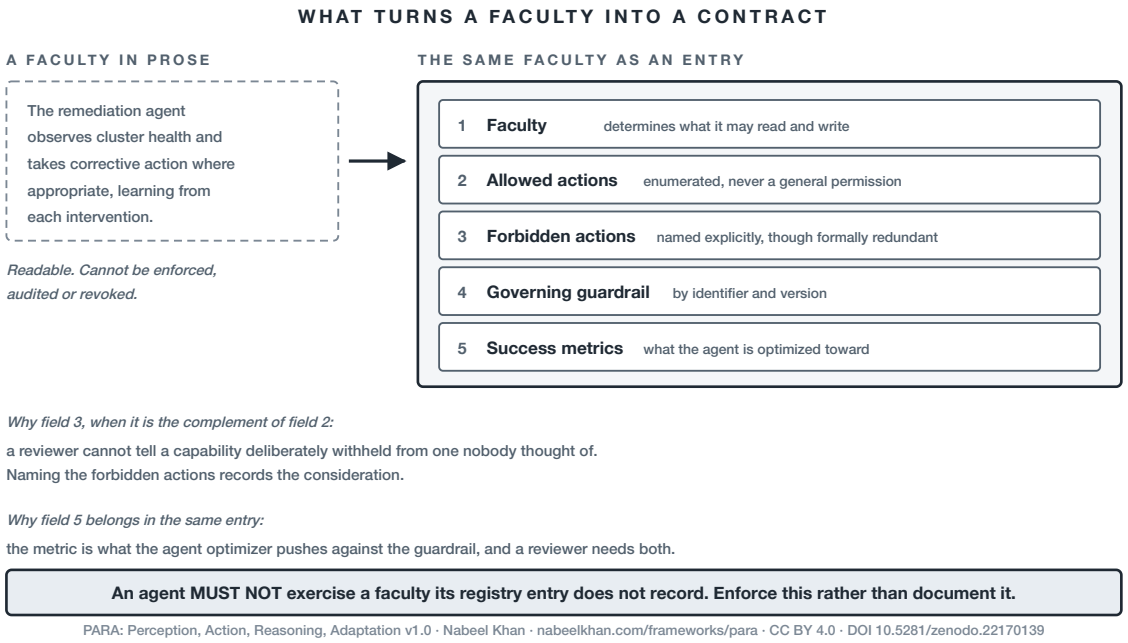


Figure 3. The registry entry that turns a faculty into a contract. Five fields: the faculty, the enumerated allowed actions, the explicitly named forbidden actions, the governing guardrail by identifier and version, and the success metrics. Forbidden actions are named although they are formally redundant, because a reviewer cannot otherwise tell a capability deliberately withheld from one nobody thought of. Success metrics are in the same entry because the metric is what the agent's optimizer pushes against the guardrail, and a reviewer needs both to judge whether the guardrail is under pressure.

7.5 Figure 3: the entry

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8. Evidence (Normative)

8.1 What the loop produces

A conformant agent MUST produce, per cycle: the observation, the plan and its justification, the action taken, and the knowledge updated. These four together are the agent's Delivery Evidence in the sense of the pattern language [1].

8.2 The justification is the part most easily lost

Implementations record observations and actions readily, because both are events. **A justification is not an event and must be deliberately written**, and an agent whose evidence contains observations and actions and no reasoning has recorded what it did and not why, which is the gap Section 1.1 describes.

8.3 Adaptations are recorded as changes, not as states

An adaptation MUST be recorded as a change with its prior state, per item 3 of 6.4, and MUST NOT be recorded only as the new state of the knowledge. A runbook that shows what it now says, and not what it said or who changed it, cannot support the review that 4.4 requires.

8.4 The BOE Declaration

An agent's evidence SHOULD be stated as a BOE Declaration [1]: the boundaries fixed are the guardrails on Action and on Adaptation; the optimizer freed is the reasoning and adaptation loop within them; the evidence is the four records above. **Stating it in that shape is what lets an agent's evidence join the institution's other evidence** rather than sitting in its own format.

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9. Limitations and What Has Not Been Tested

9.1 The ancestor is MAPE-K, and documented uses are emerging

Perception-action loops are established in platform and reliability practice, and extending such a loop with an explicit reasoning step and a persistent knowledge store is not original here. **MAPE-K is the ancestor.** Kephart and Chess published it in 2003 as the reference control model for autonomic and self-adaptive systems: Monitor, Analyze, Plan and Execute over a shared Knowledge base [6]. The correspondence is close, and it is stated here rather than left for a reader to find. Monitor is close to Perception. Analyze and Plan together are close to Reasoning. Execute is close to Action. And the Knowledge base is close to the institutional memory that Adaptation writes to.

No priority is claimed over MAPE-K, and the four-part decomposition is not claimed as original. What this specification claims is narrower. The faculties are cut by authority type rather than by activity, which is the requirement at 2.2 and is what makes a faculty something an institution can revoke. And Adaptation is bounded by the same guardrails as Action, which is the requirement at Section 6 and rests on the laundering argument at 6.3. MAPE-K is a control model, and the authority question is not the question it answers.

Documented uses of the four-part model as specified here are emerging rather than established. OP-5 carries evidence level E1 in the pattern language [1], the lowest substantive level, and this specification does not raise it.

E1 is where a published mechanism sits before an independent party has implemented and documented one. No implementation unconnected to the author has been evaluated, and version 1.0 claims no more than E1.

9.2 The laundering failure is argued, not observed

Section 6.3 describes a mechanism. **It has not been observed in a documented case**, and no study establishes that agents with unguarded adaptation authority produce degraded institutional knowledge at a measurable rate.

The argument is that the failure is undetectable downstream, which is also what makes it hard to evidence: **a failure whose signature is the absence of a second copy is difficult to count**. That is a real weakness of the argument and not only of the phenomenon, and a reader should weigh it as one.

9.3 Four faculties may be the wrong decomposition

A reader might hold that perception and reasoning are one faculty in any real implementation, that action should be split by blast radius rather than treated as one authority, or that adaptation is a governance process rather than an agent faculty. **None of those objections is answered here.**

9.4 The registry entry adds friction

Five fields per agent, maintained as agents change, is real overhead. An institution running a handful of agents may reasonably keep them in a document. **The entry earns its cost at the scale where nobody can hold the agent estate in their head**, and this specification does not state where that scale begins.

9.5 What would falsify the model

PARA is falsified if institutions operating agents with unguarded adaptation authority are found to suffer no more degradation of their institutional knowledge than institutions that guard it, measured as the rate at which runbook changes are later reversed as wrong.

That is the model's central and most contestable claim, and it is measurable without agreeing on the faculty decomposition.

Two narrower findings would refute parts of it. If agents whose reasoning is recorded are diagnosed no faster after a bad action than agents whose reasoning is not, the argument at 1.1 for naming Reasoning as a faculty fails. And if agents with registry entries acquire undeclared capabilities at the same rate as agents without them, Section 7 is documentation rather than enforcement.

9.6 Research questions

1. Do agents with unguarded adaptation authority produce runbook changes that are later reversed at a higher rate?

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2. How much faster is a bad action diagnosed when the reasoning faculty's justification was recorded?
3. How often do operational agents acquire capabilities their registry entry does not record, and by what route?
4. At what size of agent estate does the registry entry begin to pay for itself?

Anyone able to answer any of these from a real deployment is invited to make contact through the ORCID record in the front matter. Question one would move OP-5 off evidence level E1, which is the most valuable thing that could happen to this construct.

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10. Versioning

The major number changes when a change could make a previously conformant agent non-conformant: a faculty is added or removed, a registry entry field is added, or a guardrail requirement is tightened.

The minor number changes for additions and clarifications that cannot.

The four faculty names are normative and MUST NOT be renamed, and **the fourth is Adaptation**. That is stated in the front matter, at 6.1 and here, because the incorrect name has appeared in the author's own earlier material and a specification that only corrected it once would be relying on the reader to have read that once.

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11. References

Every reference was verified against a live primary source on 29 August 2026, except reference [6], verified on 30 August 2026. Each was verified with a deliberate failing control, because a check that cannot fail proves nothing.

[1] Khan, N. A. (2026). *A Pattern Language for Production LLM Platforms: Governed Routing, Agent Orchestration, and AI-Native Delivery. Pattern Language Specification.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22109864. **OP-5 is the canonical statement of this construct and governs where this document disagrees with it.**

[2] Khan, N. A. (2026). *DevOps for AI-Native Platforms: Building, Governing, and Scaling AI Infrastructure.* iSystematic Inc. ISBN 978-1-0678317-3-8. Chapter 8 is the source treatment from which this specification is drawn.

[3] Khan, N. A. (2026). *The MESA Framework: A Four-Altitude Diagnostic Model for Institutional AI Governance. Framework Specification.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22109836. Cited for the institutional altitude above which this specification does not operate.

[4] International Organization for Standardization and International Electrotechnical Commission (2023). *ISO/IEC 42001:2023, Information technology. Artificial intelligence. Management system.* First edition, December 2023.

[5] National Institute of Standards and Technology (2023). *Artificial Intelligence Risk Management Framework (AI RMF 1.0).* NIST AI 100-1, 26 January 2023.

[6] Kephart, J. O., and Chess, D. M. (2003). *The vision of autonomic computing.* Computer, volume 36, issue 1. DOI https://doi.org/10.1109/MC.2003.1160055. The origin of the MAPE-K control model, cited at 9.1 as the ancestor of this decomposition.

Appendix A. Publication Provenance

A.1 First public version of this specification. Not peer reviewed. Deposited on Zenodo, which confers a DOI, a timestamp and citability, and confers no standard status.

A.2 The construct was published before this deposit, as **OP-5** of [1] and at chapter length in [2]. **This document adds depth rather than priority**, and the priority date of the construct is that of its earlier publication rather than of this record. The four-faculty decomposition is not claimed as original either. MAPE-K, published in 2003

[6], is the ancestor of the arrangement, and 9.1 states the correspondence and what this specification claims that MAPE-K does not.

A.3 No preprint identifier exists and no preprint relation is asserted in this record's metadata.

A.4 Publication order. Prepared to publish after version 1.1 of [1], whose Section 11 names this deposit and carries a slot for its DOI.

A.5 The naming correction is recorded as a correction. Earlier material by this author renders the fourth faculty as Reflection. That is an error, it is this author's error, and 6.1 states what it costs rather than quietly using the right word from now on.

A.6 The acronym collides with an unrelated method. PARA is also the name of a personal knowledge management method, Projects, Areas, Resources, Archives, published by Tiago Forte in 2017 and unrelated to this work. The two share four letters and nothing else: that method organizes personal files, this one specifies the faculties of an autonomous agent. Where ambiguity is possible, cite the long form, Perception, Action, Reasoning, Adaptation, which is the distinguishing name.

A.7 No trademark is claimed on PARA or on the four faculty names, which is a decision rather than an omission.

A.8 The author operates a consulting practice applying this construct, which is a conflict of interest stated here.

Document History

Version	Date	Summary
1.0	30 August 2026	First public version. Four faculties by authority type, the guardrail on Adaptation and the laundering failure it prevents, the registry entry and its five fields, the acronym and loop orders, and the naming correction

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About the Author

Nabeel Khan is an enterprise AI architect and governance advisor, and Founder & Solutions Architect at iSystematic Inc. He has spent twenty-five years architecting enterprise data and machine-learning systems in government, healthcare, financial services and telecommunications across three continents, grounded in TOGAF, DMBOK, ISO 27001 and SOC 2.

He works on governed production AI; systems whose authority is bounded by the architecture, whose decisions carry the evidence needed to defend them, and whose behaviour the institution can explain afterwards. The frameworks specified here, and the regulatory knowledge behind them, are published as machine-accessible AI expertise and are callable over MCP without a key at mcp.nabeelkhan.com/api/mcp, so a reader may query this specification's contents directly and receive answers carrying their source.

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Academic citation:

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(Version 1.0). Zenodo. https://doi.org/10.5281/zenodo.22170139
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